

Question 1

Max. score: 30.00

program to check whether the given number is Armstrong number or not

print 1 if the given number is Armstrong else print 0

A positive integer is called an Armstrong number of order n if

$$abcd\dots = a^n + b^n + c^n + d^n + \dots$$

In case of an Armstrong number of 3 digits, the sum of cubes of each digits is equal to the number itself. For example:

$$153 = 1*1*1 + 5*5*5 + 3*3*3 \quad // \quad 153 \text{ is an Armstrong number.}$$

Sample input

153

Sample output

1

Python 3.8 (python 3.8.10)

```
1 n=int(input())
2 t=n
3 m=n
4 c=0
5 arm=0
6 while n>0:
7     c+=1
8     n//=10
9 while t>0:
10    r=t%10
11    arm=arm+r**c
12    t//=10
13 if m==arm:
14     print('1')
15 else:
16     print('0')
```

Question 2 !

Max. score: 30.00

Niven (Harshad) Number

print YES if given number is a Niven number or else
print NO

Sample input



Sample output



18

YES

Explanation

An integer number which is divisible by the sum of its digits is said to be a Harshad Number. An n-Harshad number is an integer number divisible by the sum of its digits. EX: 18

Python 3.8 (python 3.8.10) v

```
1 h=int(input())
2 t=n
3 s=0
4 while n>0:
5     r=n%10
6     s=s+r
7     n=n//10
8 if t%s==0:
9     print('YES')
10 else:
11     print('NO')
```


Happy Number

Print YES if given number is a Happy number or else print NO

Sample input



Sample output



32

YES

Explanation

Happy number

The happy number can be defined as a number which will yield 1 when it is replaced by the sum of the square of its digits repeatedly. If this process results in an endless cycle of numbers containing 4, then the number is called an unhappy number.

Python 3.8 (python 3.8.10)

```
1 n=int(input())
2 while n!=1 and n!=4:
3     s=0
4     while n>0:
5         r=n%10
6         s=s+r*r
7         n=n//10
8     n=s
9 if n==1:
10     print('YES')
11 else:
12     print('NO')
13
```

Automorphic number

print YES if given number is an Automorphic number
or else print NO

Sample input



Sample output



5

YES

Explanation

Automorphic Number Program in Java

Java Programs

An Automorphic number is a number whose square "ends" in the same digits as the number itself. Examples: $5 \cdot 5 = 25$, $6 \cdot 6 = 36$, $25 \cdot 25 = 625$

Python 3.8 (python 3.8.10) ▾

🔒 Auto-com

```
1 n=int(input())
2 p=n*n
3 r=p%10
4 if r==n:
5     print('YES')
6 else:
7     print('NO')
8
```

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Question 5

Max. score: 30.00

Buzz number

print YES if given number is a Buzz number or else
print NO

Sample input

1007

Sample output

YES

Explanation

A number is said to be Buzz Number if it ends with 7 or is
divisible by 7. Example: 1007 is a Buzz Number.

Python 3.8 (python 3.8.10)

```
1 n=int(input())
2 r=n%10
3 if n%7==0 or r==7:
4     print('YES')
5 else:
6     print('NO')
```

Question 6

Max. score: 30.00

Disarium number

print YES if given number is a Disarium number or else
print NO

Sample input

175

Sample output

YES

Explanation

Disarium number

A number is said to be a Disarium number when the sum of
its digit raised to the power of their respective position is
equal to the number itself.

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Python 3.8 (python 3.8.10)

```
1 h=int(input())
2 t=n
3 k=n
4 c=0
5 while n!=0:
6     c+=1
7     n//=10
8 s=0
9 while c!=0:
10    r=t%10
11    s=s+r**c
12    t=t//10
13    c-=1
14 if k==s:
15     print('YES')
16 else:
17     print('NO')
```

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twisted prime number

print YES if given number is a twisted prime number or else print NO

Sample input

97

Sample output

YES

Explanation

A number is said to be twisted prime if it is a prime number and reverse of the number is also a prime number.

Examples: 97

97 is a prime number and its reverse 79 is also a prime number.

Python 3.8 (python 3.8.10)   Aut

```
1 h=int(input())
2 rev=0
3 while n>0:
4     r=n%10
5     rev=rev*10+r
6     n//=10
7 c=0
8 i=1
9 while i<=rev:
10     if rev%i==0:
11         c+=1
12     i+=1
13 if c==2:
14     print('YES')
15 else:
16     print('NO')
```

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program to check whether the given number is perfect number or not

print "Perfect" if given number is a perfect number ,
else "not a perfect"

Perfect number is a positive integer which is equal to the sum of its proper positive divisors except itself.

For example: 6 is the first perfect number Proper divisors of 6 are 1, 2, 3 Sum of its proper divisors = $1 + 2 + 3 = 6$. Hence 6 is a perfect number.

Note : This problem follows whole numbers

Sample input



Sample output



6

Perfect

```
1  n=int(input())
2  s=0
3  i=1
4  while i<n and n!=0:
5      if n%i==0:
6          s=s+i
7      i+=1
8  if n==s and n!=0:
9      print('Perfect')
10 else:
11     print('not a perfect')
```

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Question 9

Max. score: 30.00

prime number program

check whether a given number is prime or not. if yes, then print "PRIME" else print "NOT A PRIME".

Sample input

2

Sample output

PRIME

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

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Python 3.8 (python 3.8.10)

Auto-c

```
1 n=int(input())
2 c=0
3 i=1
4 while i<=n:
5     if n%i==0:
6         c+=1
7     i+=1
8 if c==2:
9     print('PRIME')
10 else:
11     print('NOT A PRIME')
```

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Question 10

Max. score: 30.00

SPY number

print YES if given number is a SPY number or else
print NO

Sample input

1124

Sample output

YES

Explanation

A spy number is a number where the sum of its digits equals the product of its digits. For example, 1124 is a spy number, the sum of its digits is $1+1+2+4=8$ and the product of its digits is $1*1*2*4=8$.

Python 3.8 (python 3.8.10)

Auto-comp

```
1 n=int(input())
2 s=0
3 p=1
4 while n>0:
5     r=n%10
6     s=s+r
7     p=p*r
8     n//=10
9 if s==p:
10     print('YES')
11 else:
12     print('NO')
```

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Question 11

Max. score: 30.00

Python 3.8 (python 3.8.10)

Special number

print YES if given number is a Special number or else
print NO

Sample input



Sample output



145

YES

Explanation

If the sum of the factorial of digits of a number (N) is equal to the number itself, the number (N) is called a special number.

```
1 h=int(input())
2 t=n
3 sum1=0
4 while n>0:
5     rem=n%10
6     sum2=1
7     i=1
8     while i<=rem:
9         sum2=sum2*i
10        i+=1
11    sum1=sum1+sum2
12    n=n//10
13 if t==sum1:
14     print('YES')
15 else:
16     print('NO')
```

Question 12

Max. score: 30.00

Perfect Square check

print YES if given number is a perfect square or else print NO

Sample Input

10

Sample output

NO

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Python 3.8 (python 3.8.10)

```
1 n=int(input())
2 t=n
3 sq=n**0.5
4 if t==sq**2:
5     print('YES')
6 else:
7     print('NO')
```

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Question 13

Max. score: 30.00

Palindrome number

print "Palindrome" if given number is a palindrome ,
else print "Not a palindrome"

Sample input



Sample output



121

Palindrome

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Python 3.8 (python 3.8.10)

Auto-complete

```
1 h=int(input())
2 t=n
3 rev=0
4 while n>0:
5     r=n%10
6     rev=rev*10+r
7     n=n//10
8 if t==rev:
9     print('Palindrome')
10 else:
11     print('Not a palindrome')
```

Question 14

Max. score: 30.00

perfect squares in given range

print perfect squares in the given range

Sample input



Sample output



10

1 4 9

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

1 ≤ n ≤ 10⁹ for each input file

Python 3.8 (python 3.8.10)

Auto

```
1 n=int(input())
2 i=1
3 while i<=n:
4     sq=i*i
5     if sq<=n:
6         print(sq,end=' ')
7     i+=1
8
```


Question 15 

Max. score: 30.00

Python 3.8 (python 3.8.10) 

Neon number

print YES if given number is a neon number or else
print NO

Sample input



Sample output



9

YES

Explanation

A neon number is a number where the sum of digits of square of the number is equal to the number. The task is to check and print neon numbers in a range.

Example: Given $n = 9$ and sum of the digits of the

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```
1 h=int(input())
2 t=n
3 sq=n*n
4 s=0
5 while sq>0:
6     r=sq%10
7     s=s+r
8     sq//=10
9 if t==s:
10     print('YES')
11 else:
12     print('NO')
```